

BOW – Original data

- The experiment :
 - **BoW (CountVectorizer), TF-IDF**
 - **Stemming , Lemmatization**
 - With/without **Stopwords**
- models:
 - SVM
 - Multinomial Naïve Bayes
 - MLP
 - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

Model	Best Accuracy	Configuration
SVM (SVC)	0.6616	BoW + Stem + StopWords kept
Naive Bayes (MultinomialNB)	0.6738	BoW + Lemma + StopWords removed
MLP	0.5862	TF-IDF + Stem + StopWords removed
HMM	0.1699	BoW / TF-IDF + Stem + StopWords kept

SVM

No big different between: (stem,lemma) , (BoW,TF-IDF), (stopwords kept or removed)

Multinomial Naive Bayes

Remove stopwords always improves the performance

Lemma is a bit better than stem

TF-IDF does not affect the performance

MLP

LESS performance than SVM,NB

Remove stopwords always improves the performance (MLP is sensitive to noise)

Lemma is almost as Stem

Small improvement when using TF-IDF

HMM

Worst model with huge gap with other models

HMM is designed for time sequences (BOW,TF-IDF does not)

Overall Conclusion

- **Multinomial Naive Bayes** is the best-performing model for BoW-based text classification on the original dataset.
- **SVM** provides strong and stable performance with minimal sensitivity to preprocessing choices.
- **MLP** benefits from noise reduction and TF-IDF but remains less effective than SVM and NB.
- **HMM** is inappropriate for non-sequential text representations such as BoW and TF-IDF.

BOW – under sampling the high classes

- applying **undersampling** on large classes (10,000) to be close to small classes on range(9,000–5,000)
- The experiment:
 - **BoW (CountVectorizer)** , **TF-IDF**
 - **Stemming** , **Lemmatization**
 - With/without **Stopwords**
- models:
 - SVM
 - Multinomial Naive Bayes
 - MLP
 - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

Model	Best Accuracy	Configuration
SVM	0.6383	TF-IDF + Stem + Stopwords removed
MLP	0.6315	BoW + Stem + Stopwords removed
Naive Bayes	0.6013	BoW + Stem + Stopwords removed
HMM	0.1198	TF-IDF + Lemma + Stopwords kept =BoW + Stem + Stopwords kept

SVM

TF-IDF improves performance compared to BoW.

Removing stopwords gives a small but consistent improvement.

Stemming and lemmatization show very similar performance.

Overall, SVM remains stable under undersampling with minor sensitivity to preprocessing.

Multinomial Naive Bayes

Removing stopwords consistently improves performance.

Best result is achieved using BoW + Stemming + Stopwords removed.

Lemmatization performs slightly worse than stemming in the undersampled setting.

TF-IDF does not improve performance for Naïve Bayes.

MLP

MLP performs better than Naïve Bayes but still slightly below SVM.

Removing stopwords always improves results, confirming that MLP is sensitive to noise.

Stemming performs slightly better than lemmatization.

TF-IDF does not significantly improve performance over BoW in the undersampled data.

HMM

HMM remains the worst-performing model with a large gap compared to other models.

Best performance is achieved when stopwords are kept, but accuracy remains very low.

This confirms that HMM is not suitable for non-sequential text representations such as BoW and TF-IDF.

Overall Conclusion (Undersampled Data)

- SVM remains the most reliable and best-performing model after undersampling.
- MLP benefits from noise reduction but still underperforms compared to SVM.
- Multinomial Naive Bayes is negatively affected by undersampling, especially when TF-IDF is used.
- HMM is inappropriate for text classification tasks that do not preserve sequential information

So in POS will use only the stem + stopwords removed

BOW – rosrus

- Data Balancing Strategy (ROS–RUS)

To address the severe class imbalance in the original dataset, a **hybrid resampling strategy** was applied, combining **Random OverSampling (ROS)** and **Random UnderSampling (RUS)**.

- Avoids oversampling of minority classes (which may cause overfitting).
- Avoids undersampling of majority classes (which may discard useful information).
- The experiment :
 - **Bag of Words (BoW) using CountVectorizer** was used **only**
 - **Stemming , Lemmatization**
 - With/without **Stopwords**
- models:
 - SVM
 - Multinomial Naïve Bayes
 - MLP
 - HMM
- MLP/HMM is tested with dimension reduced data (TruncatedSVD)

Model	Best Accuracy	Configuration
SVM (SVC)	0.4669	BoW + Stem + StopWords removed
Multinomial NB	0.5622	BoW + Lemma + StopWords removed
MLP	0.3688	BoW + Stem + StopWords removed
HMM	0.1699	BoW + Stem + StopWords removed

SVM

Performance is **significantly lower** than undersampling and original data.

Removing stopwords slightly improves results.

Lemmatization and stemming show no major difference.

ROS–RUS introduces duplicated samples, which negatively affects margin-based classifiers

Multinomial Naive Bayes

Most robust model under ROS–RUS.

Removing stopwords consistently improves performance.

Lemmatization performs slightly better than stemming.

Handles duplicated samples better due to its probabilistic nature.

MLP

Suffers heavily from ROS–RUS.

Very sensitive to noise and duplicated data caused by oversampling.

Removing stopwords improves performance, but overall accuracy remains low.

Indicates poor generalization under this balancing strategy.

HMM

Worst-performing model.

Accuracy remains close to random guessing.

Overall Conclusion (ROS–RUS)

Multinomial Naïve Bayes is the most stable model under ROS–RUS balancing.

SVM performance degrades due to duplicated samples affecting decision boundaries.

MLP is highly sensitive to oversampling noise.

HMM remains inappropriate for BoW-based text classification.

Compared to undersampling-only and original data, **ROS–RUS is the weakest balancing strategy** for this task